

Regularized JEPAs for Training Agents Inside of World Models

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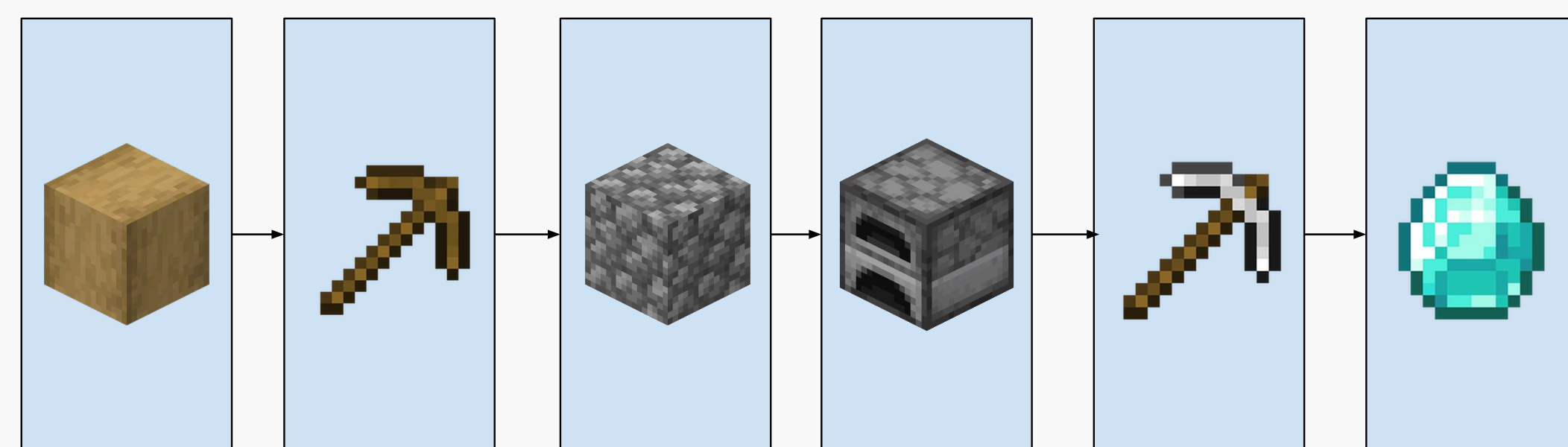
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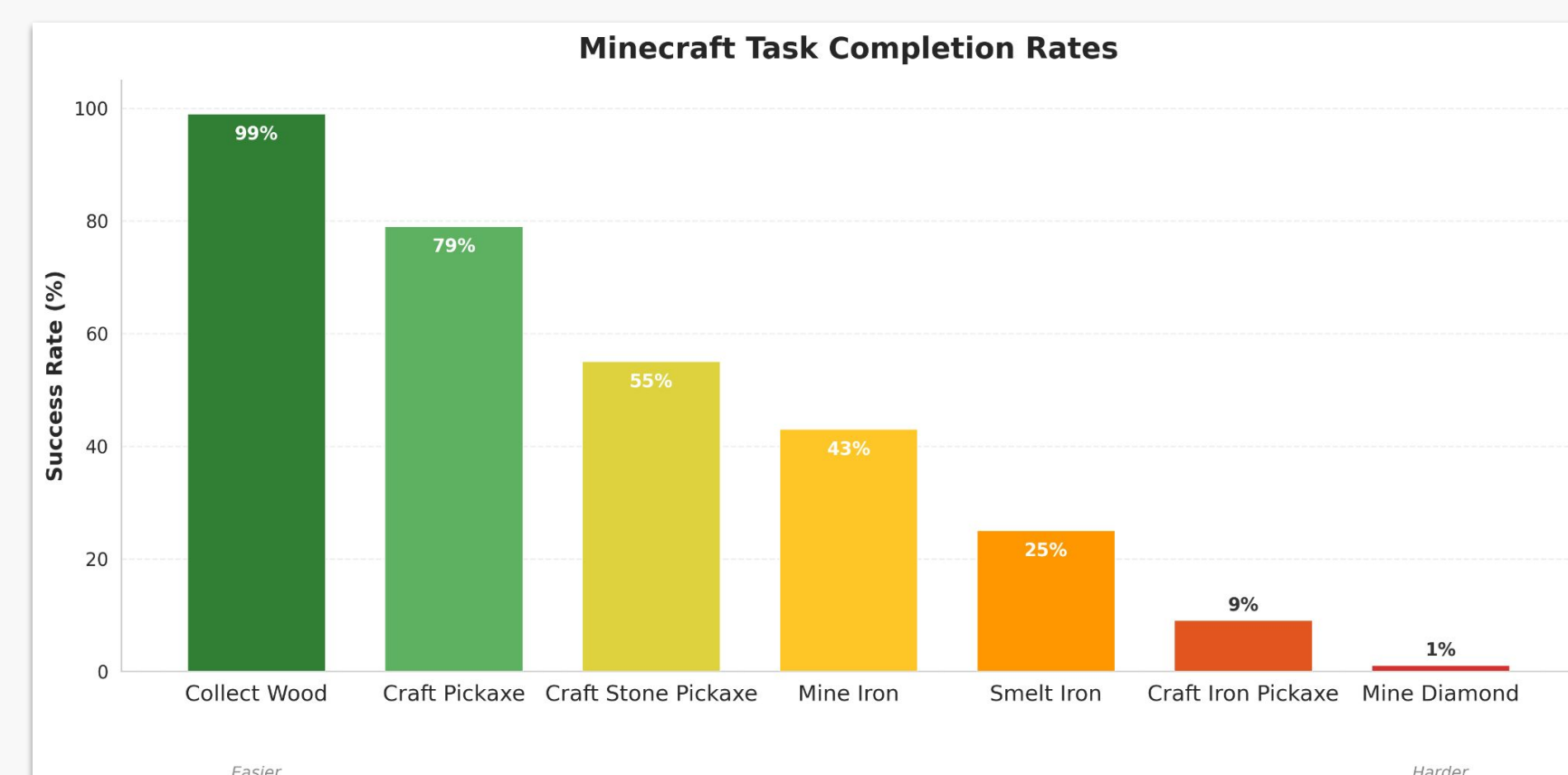
Abstract

Reinforcement learning inside of world models has recently emerged as a successful paradigm for training intelligent agents (a process known as imagination training). In the video game Minecraft, this has led to agents capable of accomplishing complex tasks like mining diamonds purely from offline training on a combination of video and environment interactions. Recent work in the area still uses pixel-wise reconstruction for training world models. Joint Embedding Predictive Architectures have been shown to improve efficiency by extracting meaningful representations for world modeling without the need for pixel-wise reconstruction. Improved regularization techniques have also led to improvements in stability of Joint Embedding Architectures by preventing representation collapse. We benchmark a regularized Joint Embedding Predictive Architecture against prior-reconstruction-based imagination training to train intelligent agents inside of the game Minecraft. Minecraft facilitates meaningful measurement of an agent's ability to interact with a complex environment to accomplish a diverse set of tasks. We evaluate the expected gains in performance and efficiency against the current state of the art architecture for the task of mining diamonds.

Introduction



- Minecraft as a proxy for agent intelligence
- Energy and resource efficient
- Complex and highly variable environment



- Tasks require many actions
- Evaluates extended planning and execution

Architecture

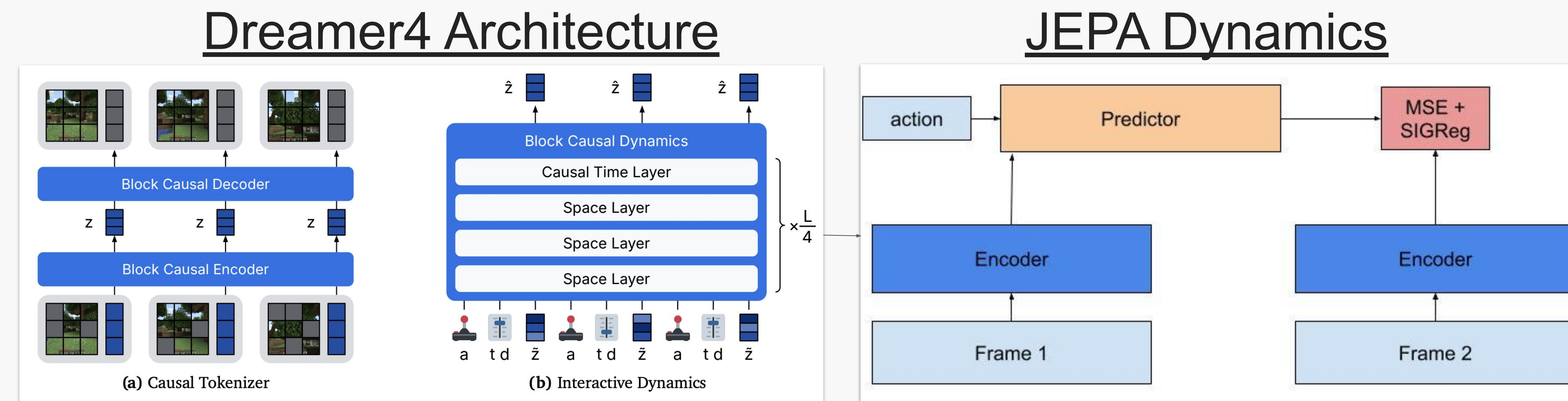
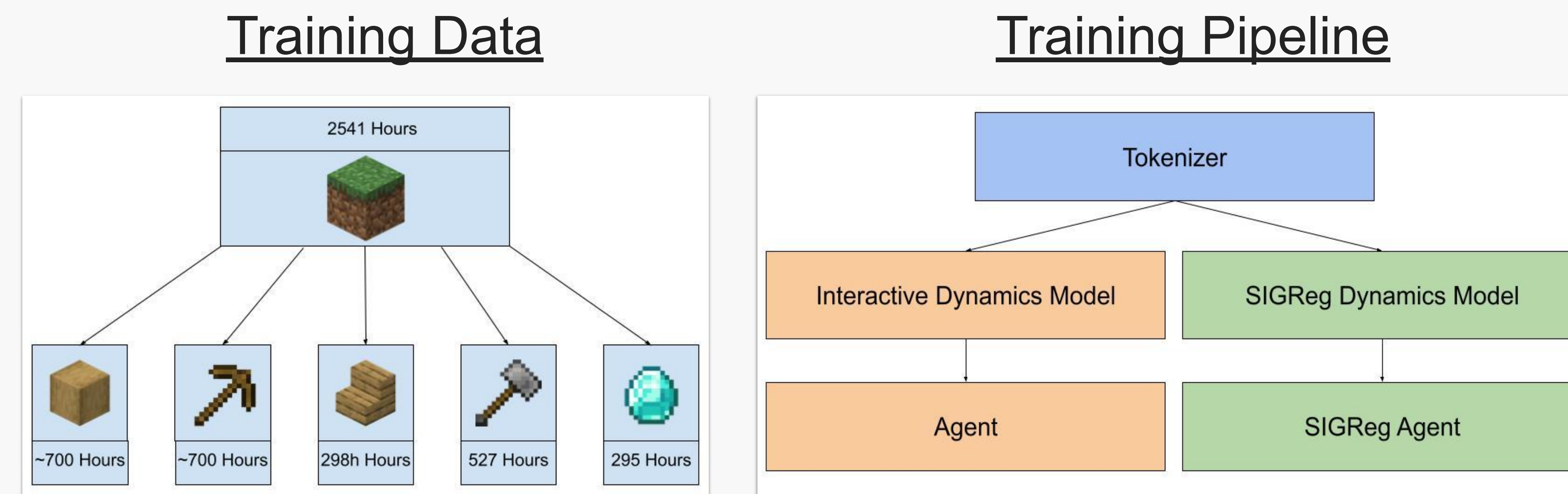


Figure taken from [1].

- We modify the dynamics model of the Dreamer4 architecture to predict the next frame's embeddings
- Joint Embedding optimization enables efficient planning and robust modeling of environment interactions by learning in representation space

Model Training and Evaluation



Original Training Pipeline

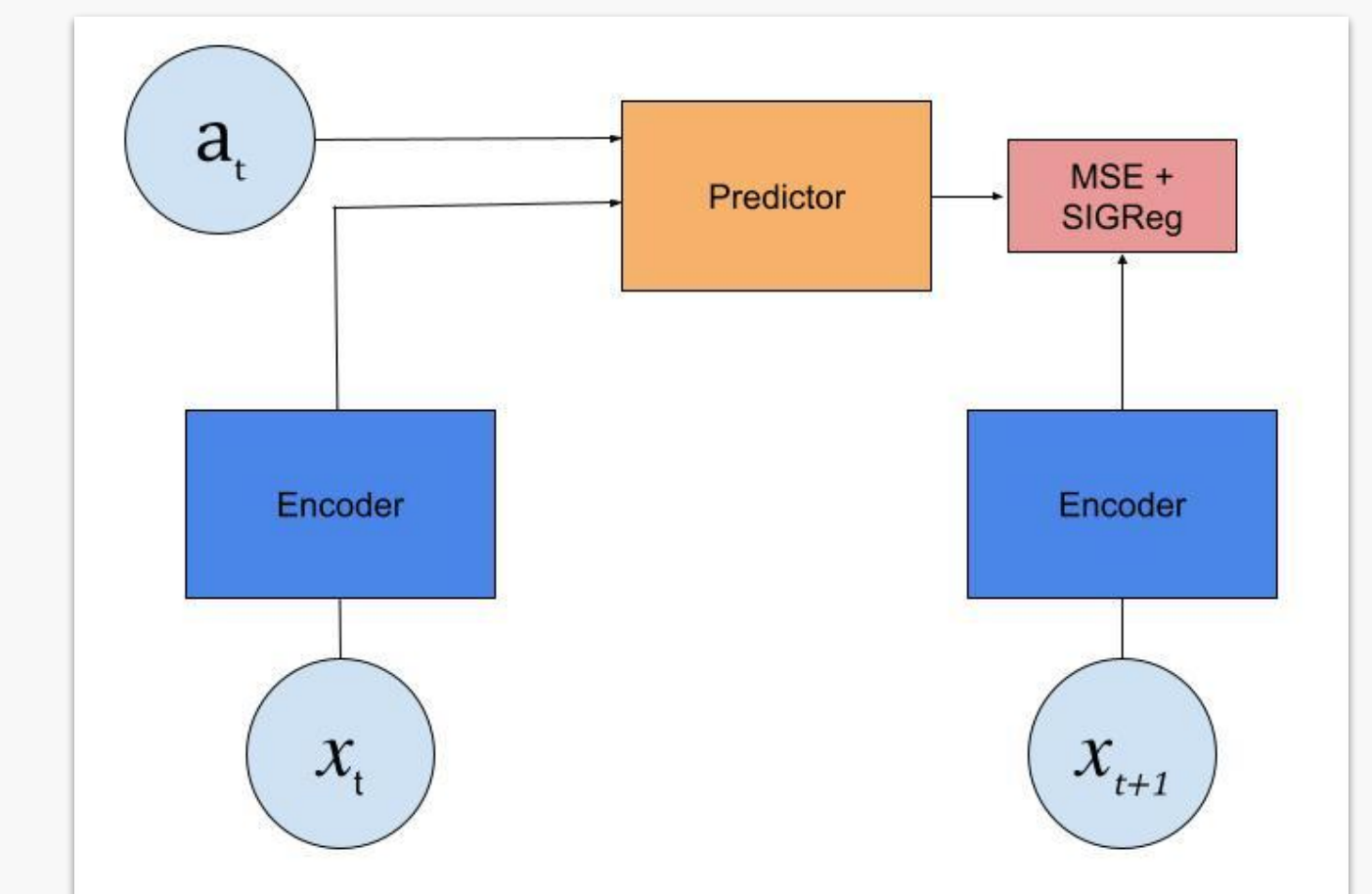
Over 400,000 hours of training



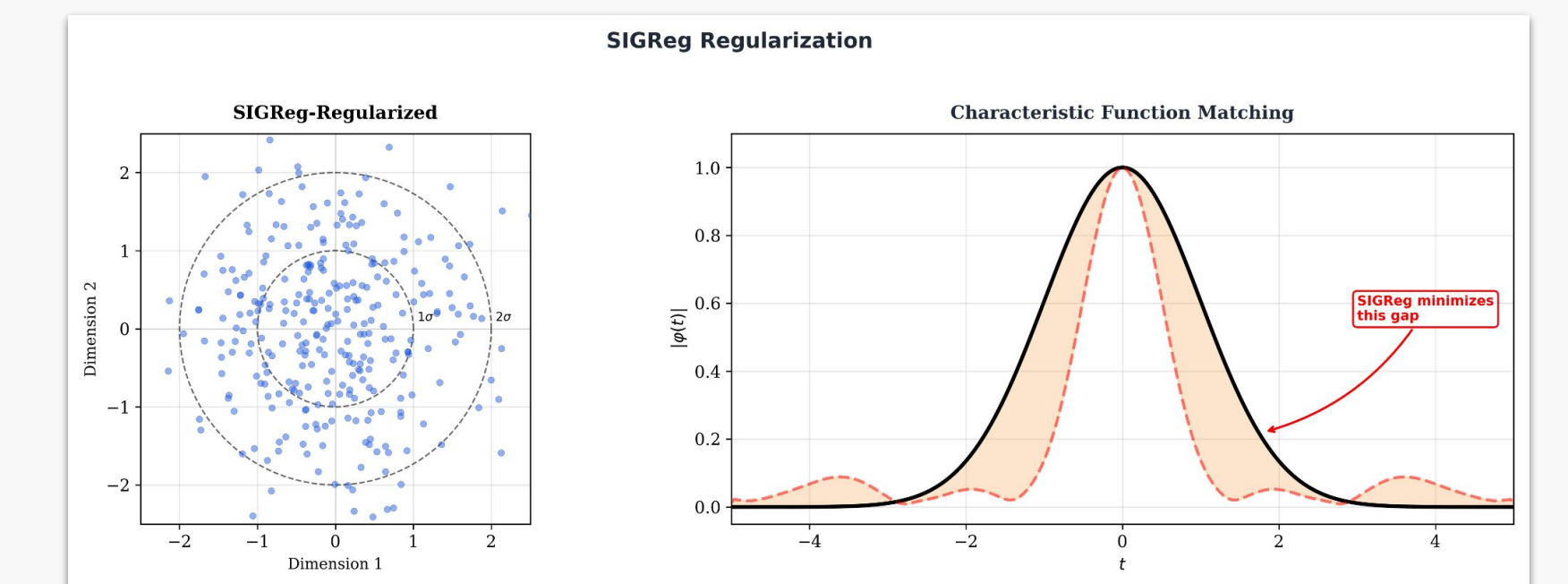
- Dreamer style models increase efficiency by removing the need for online reinforcement learning, instead training agents inside of their world models
- Joint embedding architectures have been shown in other contexts to produce better, more efficient world models than standard generative architectures

JEPAs and SIGReg

LeJEPA Optimization



- Joint Embedding Predictive Architectures learn features in representation space



- Sketched Isotropic Gaussian Regularization prevents representation collapse
- Avoids past heuristics like student-teacher and stop-gradient

Future Work

- Explore other novel architectures for world modeling from video, such as VJEPA2 and NeDreamer
- Examine alternate algorithms for training agent tokens and the tokenizer

References

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- [4] Bowen Baker, Ilge Akkaya, Peter Zhokhov, Joost Huizinga, Jie Tang, Adrien Ecoffet, Brandon Houghton, Raul Sampedro, and Jeff Clune. Video pre-training (vpt): Learning to act by watching unlabeled online videos, 2022. URL <https://arxiv.org/abs/2206.11795>.
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